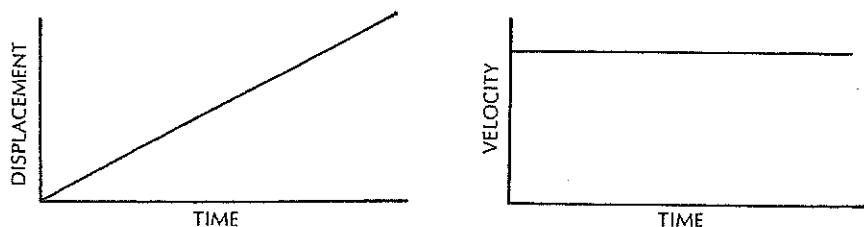
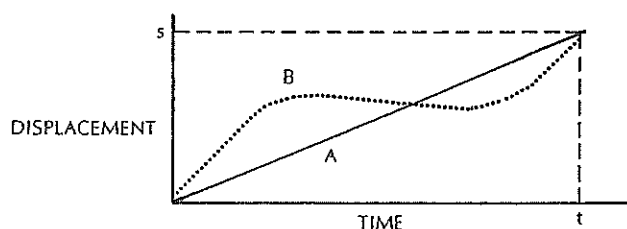


- NOTE: 1. While speed and direction remain unchanged the velocity remains the same.
 2. The graphs below represent motion with a velocity which does not vary during the considered period of time.



Velocity which does not vary is termed **CONSTANT** or **UNIFORM VELOCITY**.

3. Consider the two motions represented below.



Motions A and B both represent a displacement s in a time interval t .

$\therefore$ in each case v_{av} is the same (s/t)

Motion A is uniform (or constant) velocity while motion B is a varying velocity. Therefore for motion A, the velocity at any instant is equal to the average velocity which is equal to the uniform (or constant) velocity.

TYPE EXAMPLE 8: Determine the displacement of a body which moves with a uniform velocity of 2.4 m s^{-1} for 50 s .

$$\begin{aligned} v_{av} &= 2.4 \text{ m s}^{-1} & v_{av} &= \frac{s}{t} \\ s &= ? & \therefore s &= v_{av} \times t \\ t &= 50 \text{ s} & &= 2.4 \times 50 \\ & & &= 120 \text{ m} \end{aligned}$$

i.e. the displacement is 120 m in the direction of the velocity.

SET 6: VELOCITY I

- What is the average speed of a sprinter who runs 100 m in 12.5 s ?
- A car takes 80 s to complete one lap of a 2 km circuit. What is its average speed (in m s^{-1})?
- A body travels 36 m in 4 s . What is its average speed?
- How far will a bird fly in 25 s travelling with an average speed of 6 m s^{-1} ?
- A person swims from one end of a pool to the other at an average velocity of 1.5 m s^{-1} in 20 s . What is the length of the pool?
- Determine the distance travelled by a plane in 1 hour if it flies due south with a uniform velocity of 90 m s^{-1} .
- How long will it take for a body moving with a uniform velocity of 7 m s^{-1} to cover 98 m ?
- A cricket ball is thrown a distance of 90 m from the boundary to the wicket at an average horizontal speed of 30 m s^{-1} . How long will it take for the ball to reach the wicket?
- Determine the time taken for a horse running with an average speed of 14 m s^{-1} to complete one circuit of a 840 m track.
- How long will it take to complete a trip from Perth to Kalgoorlie, a distance of 595 km , travelling at an average speed of 70 km h^{-1} ?
- A train moving with an average speed of 80 km h^{-1} takes 0.25 hours to travel from one railway siding to the next. What is the distance between the two sidings?
- Determine the average speed (km h^{-1}) of a plane which flies a distance of 720 km in 1.5 hours ?

SET 7: VELOCITY II

- An athlete, commencing at a point due east, completes half a lap of a 440 m track in 25 s . Determine the athlete's:
 - average speed.
 - average velocity (in westerly direction).
 - average velocity if the athlete completes one full lap at the same pace.
- A car travels with an average speed of 20 m s^{-1} due north for 3 km then due west for another 4 km calculate:
 - the total time taken for the journey.
 - the displacement of the car.
 - the average velocity of the car.
- A person walks for 10 s at 1 m s^{-1} then for another 10 s in the same direction at 2 m s^{-1} . Find the:
 - total displacement.
 - average velocity.
- A body travels 200 m at a uniform velocity of 4 m s^{-1} and a further 150 m in the same direction at a uniform velocity of 7.5 m s^{-1} . Determine the:
 - total time taken.
 - average velocity for the whole journey.
- A cyclist travels at 6 m s^{-1} S for 10 minutes and then returns 600 m along the same path. What is the final displacement of the cyclist?